

AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

08

Live sessions

90 min

Per session

04 weeks

Course duration

✓ **Runs 100% on Google Colab**
Nothing to install, works even on a Chromebook

🔗 **Built for beginners**
No prior Python or machine learning required

📁 **Live code-along on Civil Engineering datasets**
Materials · Buildings & Energy · Transport · Environment · Geotechnics

📓 **Hands-on notebooks, cross-domain assignments & solutions**
Plus certificate & recordings (till 2 weeks after the course)

Course outline

01 Python for Civil Engineers
AI/ML basics, Colab, NumPy & pandas with engineering data

02 From Raw Data to ML-Ready
Explore, clean, visualize & split engineering data

03 Your First Prediction Model
Linear regression, error metrics & cross-validation

04 Trees & Ensemble Models
Decision Trees, Random Forest & boosting for nonlinear problems

05 Classification for Engineering Decisions
Logistic regression, confusion matrix, precision & recall

06 Artificial Neural Networks
Architecture, activations, training & overfitting

07 Explainable AI & Engineering Judgement
Feature importance, SHAP, and engineering interpretation

08 From Prediction to Probability
Monte Carlo, ML surrogates & probability of failure

🏗️ **Designed for Civil Engineers across disciplines** — Materials, Structures, Geotechnics, Transportation, Water & Environment.

WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty /
Practising engineers / Complete beginners to AI & ML



**Saturday,
17 October 2026**

Scan to register · qudelastudios.com →



INSTRUCTOR

Dr. Abdul Waris

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D. in Geotechnical Engineering, IIT Hyderabad

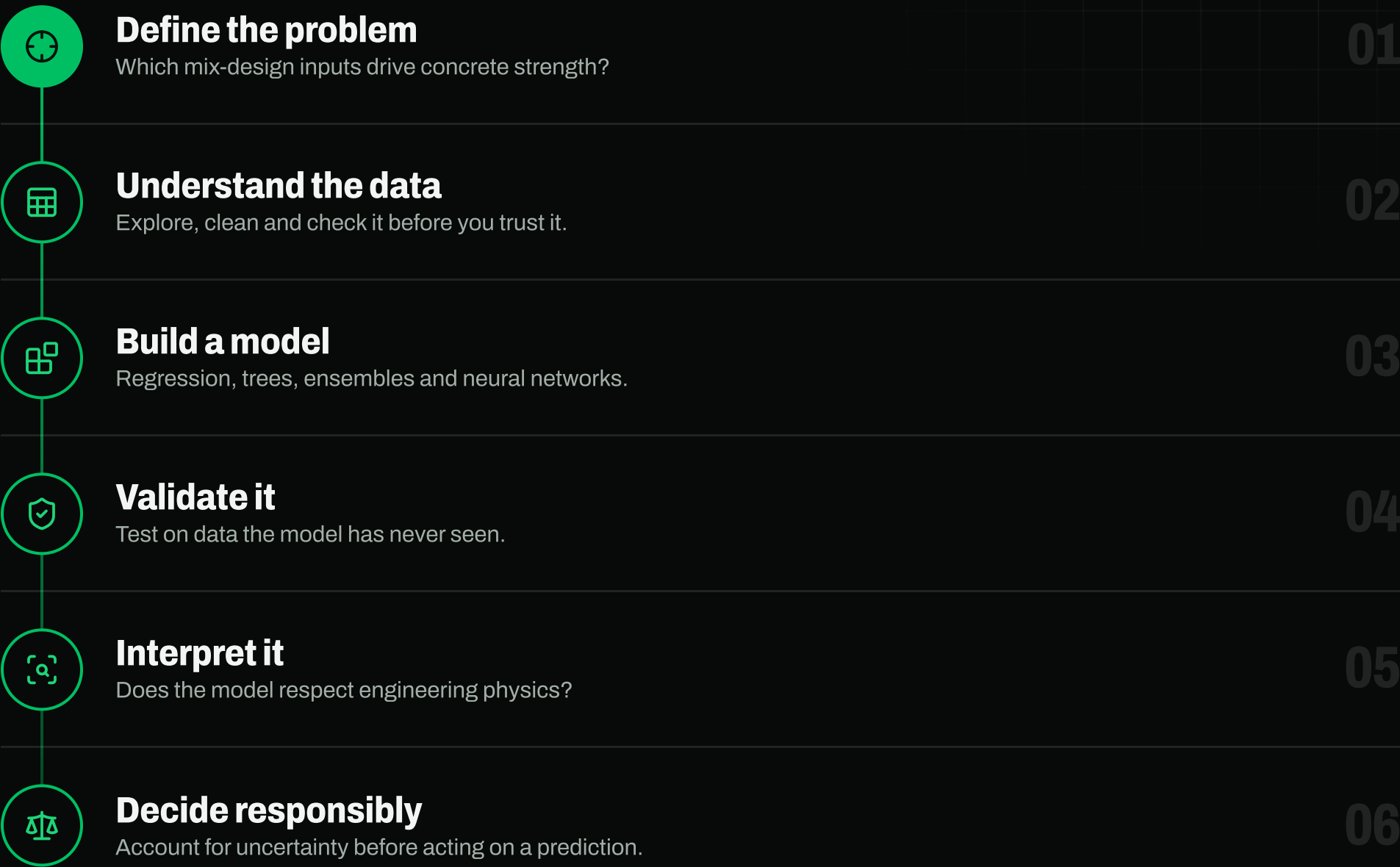
🏆 Research Excellence Award 2026, IIT Hyderabad

📖 Invited talks at Oregon State University (USA) and a
Faculty Development Programme at D.Y. Patil (Pune, India)

🎧 Resource person, five-day ML workshop at NIT Srinagar

Most ML courses start with algorithms. This one starts with engineering problems.

One ML workflow. Multiple civil engineering applications. Every session follows the workflow you would use on a real project.



DESIGNED FOR CIVIL ENGINEERS ACROSS DISCIPLINES

- Materials
- Structures
- Geotechnics
- Transportation
- Water & Environment
- Buildings & Energy

Built for civil engineers at every stage.

Whether you are still studying or already on site, you will leave able to apply ML to your own engineering problems.



UG students
B.Tech / B.E. in civil engineering



PG students
M.Tech / M.E. in any civil specialisation



PhD scholars
and early-career researchers



Faculty members
bringing data-driven methods into teaching



Practising engineers
structural, geotech, transport, water



Complete beginners
never written a line of Python? Perfect.

✔ **What you need**

- ✔ Basic civil engineering concepts
- ✔ Comfort with everyday engineering calculations
- ✔ A Google account and a web browser

✘ **What you don't**

- ✘ ~~Prior Python or programming~~
- ✘ ~~Prior machine learning experience~~
- ✘ ~~Advanced mathematics~~
- ✘ ~~Anaconda, MATLAB or any installs~~

By the end, you'll be able to...

	Frame engineering problems for ML Turn a civil engineering question into inputs, features and a target.	01
	Wrangle real engineering data Explore, clean and prepare datasets with Python, NumPy and pandas.	02
	Predict engineering quantities Build regression models for strength, response and performance.	03
	Support engineering decisions Build classifiers and judge them with precision, recall and F1.	04
	Choose the right model Compare linear, tree-based, ensemble and neural-network approaches.	05
	Validate like an engineer Catch data leakage, overfitting and unsafe extrapolation.	06
	Explain what the model learned Use feature importance and SHAP to test physical plausibility.	07
	Quantify uncertainty Estimate probability of failure with Monte Carlo and ML surrogates.	08

Sessions 01–04: the foundations.

Python, data preparation, regression and nonlinear models, on real civil engineering data.

01

Python for Civil Engineers

 **Application** · Concrete compressive strength
AI/ML basics, Colab, NumPy & pandas with engineering data.

AI/ML concepts

Google Colab

Python

NumPy

pandas

First plots

02

From Raw Data to ML-Ready

 **Application** · Concrete + cross-domain engineering data
Explore, clean, visualize & split engineering data.

EDA

Data quality

Outliers

Correlations

Train/test split

Scaling

Leakage

03

Your First Prediction Model

 **Application** · Concrete compressive strength
Linear regression, error metrics & cross-validation.

Linear regression

MAE · MSE · RMSE

R^2

Residuals

Cross-validation

04

Trees & Ensemble Models

 **Application** · Concrete + geotechnical response
Decision Trees, Random Forest & boosting for nonlinear problems.

Decision trees

Random Forest

Boosting

Model comparison



A real engineering dataset is used in every session. Hands-on examples from Materials • Structures • Buildings & Energy • Transport • Water & Environment • Geotechnical Engineering.

Sessions 05–08: prediction to decision.

05

Classification for Engineering Decisions

 **Application** · Environmental / water + geotechnical classification
Logistic regression, confusion matrix, precision & recall.

Logistic regression

Decision trees

Confusion matrix

Precision · Recall · F1

06

Artificial Neural Networks

 **Application** · Concrete & engineering response prediction
Architecture, activations, training & overfitting.

ANN architecture

Activation

Loss

Epochs

Overfitting

Convergence

07

Explainable AI & Engineering Judgement

 RESEARCH AREA

 **Application** · Geotechnical research case study
Feature importance, SHAP, and engineering interpretation.

Feature importance

Permutation importance

SHAP

Extrapolation

Physical plausibility

08

From Prediction to Probability

 RESEARCH AREA

 **Application** · Reliability / engineering exceedance
Monte Carlo, ML surrogates & probability of failure.

Monte Carlo simulation

Surrogate models

Probability of failure

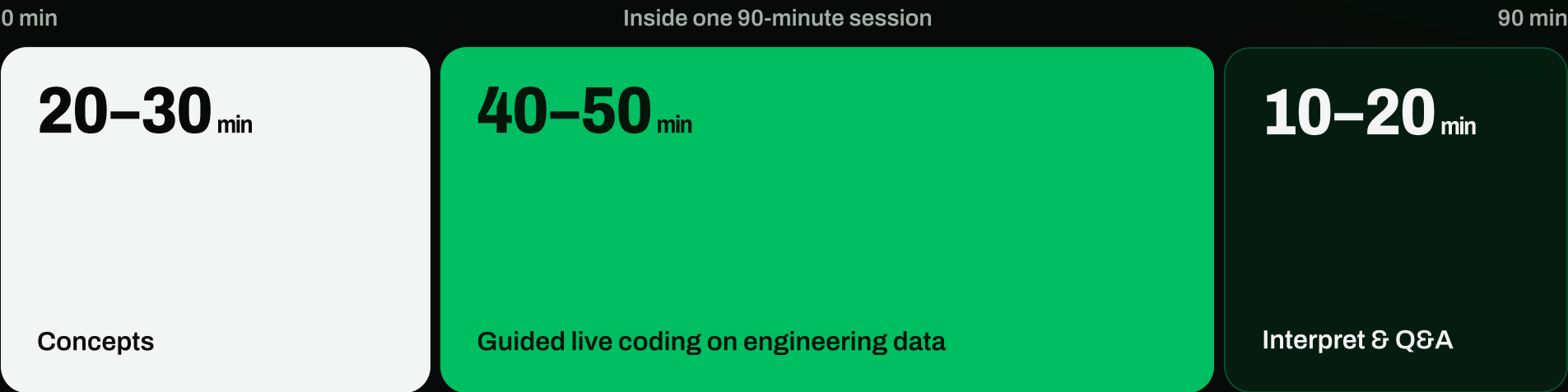
Uncertainty



Beyond a typical ML course: sessions 07 and 08 draw directly on the instructor’s research in explainable ML and reliability analysis.

Fewer slides. More code.

Every 90-minute session is live and instructor-led, and roughly half of it is spent writing code together.



Short concept explanations, then hands-on coding, then what the results mean for an engineer.

Live instructor-led coding

Code is written live with you, not pasted from slides.

Step-by-step notebooks

Every step explained in plain language inside the notebook.

Worked example, then your turn

Guided exercises follow every demonstration.

Assignments with solutions

Practise on your own, then compare with the instructor's.

Engineering discussion

Ask questions and debate what the results really mean.

Session recordings

Shared after each session; available until two weeks after the course ends.

Engineering judgement is built in. Every session discusses model limitations, validation, extrapolation and physical plausibility, not just accuracy.

Real engineering data. Right in your browser.

03_first_prediction_model.ipynb

● Google Colab · Python 3

```
[1] import pandas as pd
df = pd.read_csv("concrete_strength.csv")
df.shape

(1030, 9)
```

```
[2] from sklearn.model_selection import train_test_split
X, y = df.drop(columns="strength_mpa"), df["strength_mpa"]
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2)
```

```
[3] from sklearn.linear_model import LinearRegression
model = LinearRegression().fit(X_tr, y_tr) # your first model
```

ZERO INSTALLS

 Any laptop or Chromebook

Python

NumPy

pandas

Matplotlib

scikit-learn

SciPy

PRIMARY DATASET

UCI Concrete Compressive Strength

1,030
concrete mixes

8
input variables

0 missing values

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WHAT GOES IN

Cement

Blast-furnace slag

Fly ash

Water

Superplasticizer

Coarse aggregate

Fine aggregate

Age (days)

↓ What the model predicts

Compressive strength (MPa)

Plus hands-on examples from materials, structures, buildings & energy, transport, water & environment and geotechnical engineering.

Dr. Abdul Waris

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A computational geomechanics researcher who uses machine learning, reliability analysis and uncertainty quantification to make engineering decisions safer. His Ph.D. developed data-driven frameworks for the reliability analysis of slopes under rainfall.

- Scientific ML
- Explainable AI
- Surrogate modelling
- Reliability analysis
- Uncertainty quantification
- Offshore geotechnics



ACADEMIC JOURNEY

2016 – 2020

B.Tech, Civil Engineering

IUST, Awantipora

🏆 Ranked 2nd in a batch of 120+

2020 – 2022

M.Tech, Geotechnical Engg.

NIT Durgapur

🏆 Institute Gold Medalist

2022 – 2026

Ph.D., Geotechnical Engg.

IIT Hyderabad

🏆 PM Research Fellow · Research Excellence Award 2026

2026 – NOW

Research Associate

Universität Siegen, Germany

🎓 Teaches & supervises Master’s students

- WORKS WITH
- Python
- MATLAB
- PLAXIS
- ABAQUS
- HYDRUS
- GIS
- AutoCAD

Taught by someone who publishes this work.

10	8	9	4
international journal papers	in Q1-ranked journals	conference papers & book chapters	countries where he has presented

PUBLISHED WITH **ASCE** **Springer** **Elsevier** **Nature Portfolio**

Selected ML publications

Explainable ensemble learning for estimating strength and stiffness of clay-bearing rocks

Scientific Reports, Nature Portfolio · 2026

Reliability analysis of unsaturated soil slopes using advanced surrogate models

Journal of Computing in Civil Engineering, ASCE · in press

Detection and prediction of slope stability using interpretable machine learning

Modeling Earth Systems and Environment, Springer · 2025

Pseudo-static slope stability analysis using explainable ML techniques

Journal of Natural Hazards, Springer · 2024

Invited lectures & talks

Machine Learning for Geotechnical Engineers: Fundamentals & Practical Applications

Oregon State University, USA · Invited talk · 2024

Five-day workshop: ML & Finite Element Methods in Geotechnical Engineering

NIT Srinagar · Resource person · 2025

Unlocking Geotechnical Insights with Machine Learning

D.Y. Patil Institute of Technology, Pune · National FDP · 2025

Conference presentations

Oslo, Norway · Barcelona, Spain · 2025

Teaching experience



Master’s teaching & supervision

Universität Siegen, Germany



Course instructor, Structural Analysis

B.Tech students, University of Kashmir



Teaching assistant, Geotechnical Engg.

IIT Hyderabad & NIT Srinagar

Everything you need, **included.**

No software to buy and nothing to install. Every notebook opens straight in Google Colab.

- **8 hands-on notebooks**
Structured Colab notebooks, one per session
- **Curated datasets**
Cleaned, with data dictionaries
- **Cross-domain assignments**
Practice problems across civil disciplines
- **Solution notebooks**
Full instructor solutions to compare
- **Python & ML quick reference**
A cheat sheet to keep at your desk
- **Session recordings**
Available until two weeks after the course ends

COHORT FALL 2026 · LIMITED SEATS

Starts Saturday, 17 October 2026

-  8 live sessions
-  90 min each
-  4 weeks
-  Certificate
-  100% online

REGISTER NOW

qudelastudios.com

 Scan the code to open the course page



Seats are limited so everyone gets real support during live coding.